CONTINUOUS AND SMOOTH PLURISUBHARMONIC EXHAUSTIONS: THREEFOLD COUNTEREXAMPLES AND SURFACE REGULARIZATION

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Abstract. We construct smooth Kähler and non-Kähler threefolds with locally Lipschitz plurisubharmonic exhaustions but no C^2 plurisubharmonic exhaustion. The Kähler example is quasi-projective and arises from a continuous semipositive metric on Koike's modified Grauert example; a radius criterion excludes every C^2 exhaustion. In dimension two, we prove continuous-to-smooth regularization for divisor complements in compact Kähler surfaces, hence for smooth quasi-projective surfaces. The proof combines level averaging of positive dd^c-closed currents with the surface current theorem and Hodge index. We also give boundary existence criteria and prove holomorphic convexity for reduced pure n-dimensional spaces with a continuous psh exhaustion and a holomorphic map to $\mathbb{C}^{n-1}$ whose fibers all have dimension at most one. The general noncompact surface problem remains open here.

Contents

1. Introduction and main results 2
2. The ruled surface and its metrics 4
3. Exhaustions and the radius criterion 6
4. A non-Kähler modification over one point 7
5. Surface regularization and boundary criteria 9
6. Holomorphic convexity for curve fiberings 16
References 19

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1. Introduction and main results

A continuous plurisubharmonic exhaustion need not admit a smooth replacement, even on a quasi-projective manifold. We construct threefold examples and prove affirmative results for compactifiable surfaces and spaces fibered by curves. Throughout, psh means plurisubharmonic and smooth means C^∞.

Theorem 1.1. *Let C be the smooth projective completion of $y^2 = x^5 - x$, let $K = K_C$, and choose a nonsplit extension*

$$0 \longrightarrow K \longrightarrow V \longrightarrow \mathcal{O}_C \longrightarrow 0. \tag{1.1}$$

Set

$$\begin{aligned} S = \mathbf{P}_{\text{lines}}(V), \quad \pi : S \to C, \quad Y = \mathbf{P}(K), \\ L = \mathcal{O}_S(Y) \otimes \pi^* K, \qquad M = \operatorname{Tot}(L^{-1}). \end{aligned} \tag{1.2}$$

Then S is projective and M is a smooth quasi-projective Kähler threefold. The line bundle L admits a locally Lipschitz Hermitian metric with psh weights, but no C^2 semipositive Hermitian metric. The threefold M admits a locally Lipschitz psh exhaustion, but no C^2 psh exhaustion.

The ruled surface is Koike's modified Grauert example [1, Example 4.1]; the semipositivity obstruction is known [1, Theorem 1.4][2, Remark 2.8]. Our proof supplies an explicit continuous metric and uses the radius criterion (Corollary 3.4): for a line bundle A over a compact complex manifold and $r \in \{2, \infty\}$,

$$\begin{gathered} \operatorname{Tot}(A^{-1}) \text{ has a } C^r \text{ psh exhaustion} \\ \Longleftrightarrow \\ A \text{ has a } C^r \text{ semipositive metric.} \end{gathered}$$

The obstruction therefore applies to every C^2 exhaustion. The C^1 threshold remains undetermined.

Theorem 1.2. *For every $x \in M$ there is a smooth non-Kähler threefold N and a proper holomorphic modification $\mu : N \to M$ such that*

$$N \setminus \mu^{-1}(x) \simeq M \setminus \{x\}.$$

The threefold N admits a locally Lipschitz psh exhaustion, but no C^2 psh exhaustion.

Hironaka's opposite-order blowups produce the non-Kähler configuration over x. Proposition 4.1 transfers the exhaustion properties through this compactly supported modification.

Theorem 1.3. *Let S be a smooth compact connected Kähler surface and $D \neq 0$ an effective divisor. If $S \setminus \operatorname{supp} D$ admits a continuous psh exhaustion, it admits a smooth one. Consequently, the same implication holds for every smooth complex quasi-projective surface.*

The proof averages positive dd^c-closed currents over exhaustion levels to obtain a current with zero Lelong numbers. The current theorem of Dinh–Nguyên–Sibony [15] and Hodge index then give the following alternative: either $S \setminus \operatorname{supp} D$ is a modification of a Stein space, or a divisor $A = \sum a_i D_i$, $a_i \in \mathbb{Z}_{>0}$, satisfies

$$\operatorname{supp} A = \operatorname{supp} D, \qquad A^2 = 0, \qquad \mathcal{O}_S(A) \text{ is unitary flat near } D.$$

Its canonical section yields a pluriharmonic logarithmic collar. No normal-crossing or minimality hypothesis is imposed on S, D. Section 5 also gives separate numerical and neighborhood criteria for existence of an exhaustion.

Theorem 1.4. *Let X be a reduced complex space of pure dimension $n \geq 2$ with a continuous real-valued psh exhaustion. If a holomorphic map*

$$F : X \longrightarrow \mathbb{C}^{n-1}$$

has every nonempty fiber of dimension at most one, then X is holomorphically convex and admits a smooth psh exhaustion.

The fiber bound includes exceptional fibers. Neither properness of F nor normality of X nor a restriction on $\operatorname{Sing} X$ is required. Analytic fiber cycles and Bishop compactness establish the stability needed for Vîjîitu's convexity criteria [17]. For irreducible surfaces, convexity already follows from [4, Theorem 1.1].

These regularization results assert existence of a smooth exhaustion, not approximation of the prescribed one. They do not settle the problem for arbitrary noncompact complex surfaces. Within smooth quasi-projective manifolds, Theorems 1.1 and 1.3 place the least possible counterexample dimension at three. The separate elliptic-neighborhood criterion uses the September 2026 preprint [11]; the main surface theorem does not depend on that input.

Conventions. We use

$$d^c = \frac{i}{4\pi}(\bar\partial - \partial), \qquad dd^c = \frac{i}{2\pi}\partial\bar\partial, \qquad \|e\|_h^2 = e^{-\varphi}, \qquad \Theta_h(A) = dd^c\varphi.$$

For local frames $e_j = g_{ij} e_i$,

$$\varphi_j = \varphi_i - \log |g_{ij}|^2. \tag{1.3}$$

A continuous Hermitian metric is semipositive when its weights are psh. For C^2 functions, write $\mathcal{L}_f(\xi) = \sum f_{z_j \bar z_k} \xi_j \overline{\xi_k}$. For the canonical section s_D,

$$dd^c \log \|s_D\|_h^2 = [D] - \Theta_h(\mathcal{O}(D)). \tag{1.4}$$

An exhaustion is a continuous real-valued function with compact sublevel sets. Continuous or smooth weak completeness means existence of a psh exhaustion of that regularity. On complex spaces, smoothness means local restriction of smooth ambient functions. The projectivization $\mathbf{P}_{\text{lines}}(E)$ parametrizes lines, with tautological bundle $\mathcal{O}_{\mathbf{P}(E)}(-1)$.

2. The ruled surface and its metrics

2.1. **Geometry and the first-order obstruction.** The map $C \to \mathbf{P}^1$ defined by $y^2 = x^5 - x$ has six simple branch points, including infinity. Hence

$$2g(C)-2 = -4+6 = 2, \qquad \deg K = 2, \qquad \operatorname{Ext}^1(\mathcal{O}_C, K) = H^1(C, K) \simeq \mathbb{C}.$$

Choose a nonzero extension class as in (1.1). All such classes give isomorphic middle terms after scalar automorphisms of the end terms. By GAGA, V is algebraic; thus $S = \mathbf{P}_{\text{lines}}(V)$ is projective. For $Y = \mathbf{P}(K)$ and $L = \mathcal{O}_S(Y) \otimes \pi^* K$,

$$N_{Y/S} \simeq \operatorname{Hom}(K, V/K) \simeq K^{-1}, \qquad Y^2 = -2, \qquad L|_Y \simeq \mathcal{O}_Y. \tag{2.1}$$

The last trivialization is induced by $K^{-1} \otimes K$. If F is a fiber and $D \neq Y$ is an irreducible horizontal curve,

$$L \cdot Y = 0, \quad L \cdot F = 1, \quad L^2 = 2, \quad L \cdot D = Y \cdot D + 2\deg(\pi|_D) > 0.$$

Thus L is nef and big. This is the modified Grauert example of [1, Example 4.1].

Choose local frames k_i of K and lifts v_i of $1 \in \mathcal{O}_C$, with

$$k_j = g_{ij} k_i, \qquad v_j = v_i + c_{ij} k_i.$$

In the coordinates $[k_i + w_i v_i]$ near Y, and the frames $\ell_i = d_i \otimes \pi^* k_i$ with $s_Y = w_i d_i$, one obtains

$$w_j = \frac{g_{ij} w_i}{1 - c_{ij} w_i}, \qquad \ell_j = (1 - c_{ij} w_i)\ell_i. \tag{2.2}$$

Proposition 2.1. *The line bundle L has no C^2 semipositive Hermitian metric.*

Proof. Suppose its local weights φ_i are C^2 and psh. Their restrictions to Y agree and are subharmonic, hence constant. After rescaling, $\varphi_i(z, 0) = 0$. Positivity of the Levi matrix and $\varphi_{i,z\bar{z}}(z, 0) = 0$ imply $\varphi_{i,w\bar{z}}(z, 0) = 0$. Thus $a_i(z) = \varphi_{i,w}(z, 0)$ is holomorphic. Differentiating the metric transition law gives

$$\varphi_j(z, w_j) = \varphi_i(z, w_i) - \log|1 - c_{ij} w_i|^2 \quad \Longrightarrow \quad g_{ij} a_j = a_i + c_{ij}.$$

Consequently $v_i - a_i k_i$ glue to a holomorphic lift of 1, splitting the chosen extension. This contradiction is the first-order obstruction of [1, Theorem 1.4]; the calculation explicitly uses only C^2 regularity. □

2.2. **Cusp correction and a global metric.**

Lemma 2.2 (Cusp correction)**.** *Let D be a compact smooth curve in a complex surface, with $\deg N_{D/X} < 0$, and let $A|_D$ be unitary flat. Choose smooth metrics h_0 on A and h_D on $\mathcal{O}_X(D)$ such that $h_0|_D$ is flat and $\Theta_{h_D}|_{TD} < 0$. Put $q = \|s_D\|^2_{h_D}$. For $0 < a \leq \frac{1}{2}$, the local weights*

$$\varphi_0 + \Lambda q^a$$

are psh near D, strictly psh off D, and locally $C^{0,2a}$, for a suitable $\Lambda > 0$. One may take $\Lambda = 1$ when $a < \frac{1}{2}$; a sufficiently large Λ works when $a = \frac{1}{2}$.

Proof. The required metrics exist because negative degree gives a negative curvature metric on $N_{D/X}$, and metrics on restricted bundles extend by extending their logarithmic ratios to fixed smooth metrics. In a finite coordinate cover of D, write

$$D = \{w = 0\}, \quad q = |w|^2 e^{\beta(z,w)}, \quad \beta_{z\bar{z}}(z,0) > 0, \quad \tau = \log q.$$

For $w \neq 0$ and $v = (\xi, w\nu)$,

$$\begin{aligned} q^{-a}\mathcal{L}_{q^a}(v) &= a\mathcal{L}_\tau(v) + a^2|\partial\tau(v)|^2 \\ &= a\beta_{z\bar{z}}(z,0)|\xi|^2 + a^2|\nu + \beta_z(z,0)\xi|^2 + O(|w|)(|\xi|^2 + |\nu|^2). \end{aligned} \tag{2.3}$$

The leading form is positive definite. Compactness and $q^a \asymp |w|^{2a}$ therefore give, uniformly near D,

$$\mathcal{L}_{q^a}(\xi, w\nu) \geq c|w|^{2a}(|\xi|^2 + |\nu|^2).$$

Flatness of $h_0|_D$ gives $\varphi_{0,z\bar{z}}(z,0) = 0$; smoothness then yields

$$|\mathcal{L}_{\varphi_0}(\xi, w\nu)| \leq C|w|(|\xi|^2 + |\nu|^2).$$

Hence

$$\mathcal{L}_{\varphi_0 + \Lambda q^a}(\xi, w\nu) \geq (\Lambda c|w|^{2a} - C|w|)(|\xi|^2 + |\nu|^2). \tag{2.4}$$

For $a < \frac{1}{2}$, shrink the neighborhood with $\Lambda = 1$; for $a = \frac{1}{2}$, choose $\Lambda > C/c$. This proves strict positivity off D. Since $q^a = |w|^{2a}e^{a\beta}$ is locally $C^{0,2a}$, the weights extend continuously across D; the removable singularity theorem makes the extension psh. The correction is globally defined, so the metric transition law is preserved. □

Choose a positive metric h_K on K and extend h_K^{-1} on $\mathcal{O}_S(Y)|_Y$ to a smooth metric h_Y on $\mathcal{O}_S(Y)$. Set

$$h_0 = h_Y \otimes \pi^* h_K, \qquad \theta_0 = \Theta_{h_0}(L), \qquad q = \|s_Y\|^2_{h_Y}. \tag{2.5}$$

Then $h_0|_Y$ is flat and $\Theta_{h_Y}|_{TY} = -\Theta_{h_K} < 0$. Choose a, Λ as in Lemma 2.2, and $0 < \varepsilon < Q = \max_S q$ such that $\varphi_0 + \Lambda q^a$ is psh on $\{q < \varepsilon\}$. Let

$$\begin{aligned} &B > \max_{\varepsilon \leq t \leq Q} (\Lambda t^a - \log t), \\ &\chi(0) = 0, \qquad \chi(t) = \max\{\Lambda t^a, \log t + B\} \quad (t > 0), \end{aligned} \tag{2.6}$$

and define

$$h = h_0 e^{-\chi(q)}. \tag{2.7}$$

Proposition 2.3. *The metric h is continuous and semipositive, with locally $C^{0,2a}$ weights. In particular it is locally Lipschitz for $a = \frac{1}{2}$.*

Proof. On $S \setminus Y$, Poincaré–Lelong gives

$$\theta_0 + dd^c \log q = \pi^* \Theta_{h_K}(K) \geq 0. \tag{2.8}$$

On $\{0 < q < \varepsilon\}$ the weights are maxima of the psh cusp and logarithmic branches. The logarithmic branch strictly dominates on $\{q \geq \varepsilon\}$ and on a collar of its boundary, so (2.8) proves positivity there. Near Y only the cusp branch is active. Thus the weights are continuous and psh everywhere. They are $C^{0,2a}$ near Y and maxima of smooth functions away from Y. □

Remark 2.4. For $a = \frac{1}{2}$, $T = \theta_0 + dd^c\chi(q)$ has locally Lipschitz potentials and minimal singularities. Indeed, any positive current $\theta_0 + dd^c v \in c_1(L)$ has a global quasi-psh potential v, and compactness gives $v \leq \chi(q) + C_v$. This answers the bounded-potential question of [3, Section 4] for Koike's modified example; no identification with Grauert's original line bundle is asserted.

3. Exhaustions and the radius criterion

Lemma 3.1. *If $A \to Z$ is a holomorphic line bundle over a compact complex manifold and h is a continuous semipositive Hermitian metric, then $\rho(v) = \|v\|^2_{h^{-1}}$ is a continuous psh exhaustion of $\mathrm{Tot}(A^{-1})$. It inherits the local Hölder or Lipschitz regularity of h.*

Proof. For $v = \lambda_i e_i^*$ and $\|e_i\|_h^2 = e^{-\varphi_i}$,

$$\rho = |\lambda_i|^2 e^{\varphi_i}. \tag{3.1}$$

Off the zero section, $\log \rho$ is psh, hence so is its exponential. Continuity and removability extend this across the zero section. Compactness of Z makes ρ comparable to a fixed smooth squared norm; its closed sublevels are therefore compact disk subsets. The regularity follows from (3.1). □

Lemma 3.2 (Compact-group averaging)**.** *If a compact group G acts continuously on a locally compact Hausdorff space and F is a continuous exhaustion, then $\overline{F}(x) = \int_G F(gx)\, dg$ is an exhaustion. Averaging over a smooth holomorphic compact Lie group action preserves C^r plurisubharmonicity, $r \in \{2, \infty\}$.*

Proof. For $K_c = \{F \leq c\}$,

$$\{\overline{F} \leq c\} \subset G \cdot K_c.$$

Indeed, outside the compact saturation the orbitwise minimum of F exceeds c. The left side is closed, hence compact. Differentiation under the integral and averaging Levi forms prove the remaining assertions. □

Proposition 3.3 (Radius criterion)**.** *If $\mathrm{Tot}(A^{-1})$ admits a C^r psh exhaustion, where A is a holomorphic line bundle over a compact complex manifold and $r \in \{2, \infty\}$, then A admits a C^r semipositive Hermitian metric.*

Proof. Average the exhaustion over fiberwise circle rotations. In a local frame the resulting invariant function has the form

$$F(z, \lambda_i) = U_i(z, t), \quad t = \log|\lambda_i|^2, \qquad V_i(z, \zeta) = U_i(z, \mathrm{Re}\,\zeta) = F(z, e^{\zeta/2}).$$

The function V_i is psh; thus $U_{i,tt} \geq 0$. For each z,

$$U_i(z, -\infty) = F(z, 0), \qquad U_i(z, +\infty) = +\infty.$$

Convexity implies $U_{i,t} \geq 0$, and $U_{i,t} > 0$ whenever $U_i > F(z, 0)$: a zero derivative forces constancy on the preceding half-line. Fix $c > \max_Z F|_Z$. There is a unique C^r function σ_i such that

$$U_i(z, \sigma_i(z)) = c, \qquad U_{i,t}(z, \sigma_i(z)) > 0. \tag{3.2}$$

Twice differentiating gives

$$0 = U_{i,z_j\bar{z}_k} + U_{i,z_jt}\sigma_{i,\bar{z}_k} + U_{i,t\bar{z}_k}\sigma_{i,z_j} + U_{i,tt}\sigma_{i,z_j}\sigma_{i,\bar{z}_k} + U_{i,t}\sigma_{i,z_j\bar{z}_k}.$$

Since the ζ derivatives of V_i introduce factors $1/2$,

$$0 \leq \mathcal{L}_{V_i}(\xi, 2\partial\sigma_i(\xi)) = -U_{i,t}\mathcal{L}_{\sigma_i}(\xi). \tag{3.3}$$

Thus $\psi_i = -\sigma_i$ is psh. If $e_j = g_{ij}e_i$, then $\lambda_j = g_{ij}\lambda_i$, whence

$$\sigma_j = \sigma_i + \log|g_{ij}|^2, \qquad \psi_j = \psi_i - \log|g_{ij}|^2.$$

The weights ψ_i define the required metric. □

Corollary 3.4. *For $r \in \{2, \infty\}$ and a line bundle A over a compact complex manifold,*

$$\mathrm{Tot}(A^{-1}) \text{ admits a } C^r \text{ psh exhaustion}$$
$$\Longleftrightarrow$$
$$A \text{ admits a } C^r \text{ semipositive metric.}$$

Proof. Combine Proposition 3.3 with (3.1). □

Proof of Theorem 1.1. The total space has the projective compactification

$$M = \mathbf{P}_{\text{lines}}(\mathcal{O}_S \oplus L^{-1}) \setminus \mathbf{P}_{\text{lines}}(L^{-1}), \qquad v \longmapsto [(1, v)]. \tag{3.4}$$

It is therefore smooth, quasi-projective and Kähler. With $p : M \to S$, the metric of Proposition 2.3 gives the exhaustion

$$\rho(v) = \|v\|^2_{h_0^{-1}} \exp\big(\chi(q(p(v)))\big). \tag{3.5}$$

Take $a = \frac{1}{2}$ to obtain local Lipschitz regularity. A C^2 psh exhaustion would give a C^2 semipositive metric on L by Proposition 3.3, contradicting Proposition 2.1. □

4. A non-Kähler modification over one point

Proposition 4.1. *Let $f : X' \to X$ be a proper surjective holomorphic map of nonempty complex manifolds, biholomorphic outside $f^{-1}(K)$ for some compact $K \subset X$. For $r \in \{0, 2, \infty\}$, X' admits a C^r psh exhaustion if and only if X does.*

Proof. Pullback preserves psh, regularity and compact sublevels. Conversely, let F exhaust X', choose $K \subset U \Subset X$ and $c > \max_{f^{-1}(\overline{U})} F$, and take a smooth convex nondecreasing χ vanishing on $(-\infty, c]$ and tending to infinity. Then

$$G = \chi \circ F \circ f^{-1} \quad \text{on } X \setminus K, \qquad G = 0 \quad \text{on } K$$

is C^r and psh, since it vanishes on U. For each $a \geq 0$, choose b with $\chi(t) \leq a \Rightarrow t \leq b$. The closed set

$$\{G \leq a\} \subset K \cup f(\{F \leq b\})$$

is compact. The case $K = \varnothing$ is immediate. □

Proposition 4.2. *For every point x of a connected smooth complex threefold X, there is a connected smooth non-Kähler threefold N and a proper modification $\mu : N \to X$ that is biholomorphic over $X \setminus \{x\}$.*

Proof. Blow up x:

$$b : W = \mathrm{Bl}_x X \to X, \qquad E = b^{-1}(x) \simeq \mathbf{P}^2.$$

In E, the curves

$$A = \{w = 0\}, \qquad B = \{uv = w^2\}$$

meet transversely at $p = [1 : 0 : 0]$ and $q = [0 : 1 : 0]$. Over $W \setminus \{q\}$ blow up A and then the strict transform of B; over $W \setminus \{p\}$ use the reverse order. On the overlap the centers are disjoint, so the two blowups commute canonically. Gluing gives $\nu : N \to W$ and $\mu = b \circ \nu$. All centers are smooth. Points with distinct images separate in W; points with the same image lie in a common Hausdorff chart over one of the two base opens. Thus N is a smooth Hausdorff manifold. Properness is local on the target, the charts are connected and overlap, and all centers lie over x.

We verify the non-Kähler obstruction, as in [13, Section 6] and [14, Tag 08J1]. Let D_A, D_B be the exceptional surfaces dominating A, B, and let $\alpha, \beta \in H_2(N, \mathbb{R})$ be their general fiber classes. These surfaces are compact and smooth; locally they are ruled surfaces or their blowups at a point. Their fiber divisors are linearly equivalent because $A \simeq B \simeq \mathbf{P}^1$. Over p and q the respective orders give

$$\alpha = [R_p] + [T_p], \quad \beta = [T_p], \qquad \beta = [R_q] + [T_q], \quad \alpha = [T_q],$$

where every displayed component is a smooth rational curve of multiplicity one. To check the multiplicities at p, use $A = \{y = z = 0\}$, $B = \{x = z = 0\}$. In the first blowup chart $z = yt$, the first exceptional divisor is $y = 0$ and the second center is $x = t = 0$. The second blowup restricts to the surface blowup of $(x, t) = (0, 0)$; the pullback of the fiber $x = 0$ consists of its strict transform and the exceptional curve, each once. Consequently

$$[R_p] + [R_q] = 0 \quad \text{in } H_2(N, \mathbb{R}). \tag{4.1}$$

The relations take place on compact surfaces, so they hold in ordinary homology even when N is noncompact. A Kähler form would give

$$0 = \int_{R_p} \omega + \int_{R_q} \omega > 0,$$

a contradiction. □

Proof of Theorem 1.2. Apply Proposition 4.2 to M at x. The pullback $\rho \circ \mu$ is a locally Lipschitz psh exhaustion: μ is locally Lipschitz and proper. A C^2 psh exhaustion on N would descend after truncation by Proposition 4.1, contradicting Theorem 1.1. □

5. Surface regularization and boundary criteria

Throughout this section, $X = S \setminus \operatorname{supp} D$, where S is a smooth compact connected complex surface and $D \neq 0$ is effective. A *divergent collar* is a psh function on a punctured neighborhood of $\operatorname{supp} D$ tending uniformly to $+\infty$ at the boundary. An exhaustion of X always has this divergence property.

5.1. **Boundary tools.**

Lemma 5.1. (i) *If a connected component of* $\operatorname{supp} D$ *has negative definite intersection matrix, X has no continuous psh exhaustion.*

(ii) *Smooth divergent psh collars at all connected boundary components give a smooth psh exhaustion of X.*

(iii) *For a connected reduced SNC divisor $E = \bigcup E_i$, let λ be the largest eigenvalue of $I = (E_i \cdot E_j)$. If $\lambda > 0$, E has a smooth strictly psh divergent collar. If $\lambda = 0$, $I \leq 0$,* $\dim \ker I = 1$, *and*

$$A = \sum_i a_i E_i, \qquad a_i \in \mathbb{Z}_{>0}, \qquad A \cdot E_i = A^2 = 0. \tag{5.1}$$

When S is projective, A is nef.

Proof. For (i), Grauert's theorem [6, §4, subsection 8(e)] contracts the component to a point p of a normal surface. A continuous psh function on the punctured neighborhood is bounded above near p: choose a local finite projection to a bidisc with p the only point over 0. Its fiberwise maximum is continuous and psh on the punctured bidisc; across the branch locus this follows by local boundedness and removability. The maximum on $\{|z_1| \leq r,\ |z_2| = r\}$ bounds the function on vertical discs with $z_1 \neq 0$, and continuity gives the same bound at $z_1 = 0$ away from the origin. This contradicts divergence at p.

For (ii), choose disjoint neighborhoods W_E of the boundary components, with closures inside the collar domains. Choose $T_E > \sup_{\partial W_E} \Phi_E$ and a smooth convex nondecreasing χ_E, zero on $(-\infty, T_E]$ and unbounded above. Extend $\chi_E(\Phi_E)$ by zero outside W_E and sum. The sum is smooth, psh, and proper.

For (iii), Perron–Frobenius applied to $I + c\mathrm{Id}$ shows that λ is simple and has a strictly positive eigenvector. When $\lambda > 0$, rational approximation

gives $A = \sum a_i E_i$ with $a_i > 0$ and $A \cdot E_i > 0$ for all i. Thus $\mathcal{O}_S(A)|_E$ is ample. A projective embedding by a power gives a Fubini–Study metric on this restriction. Extend its weights locally to S and patch them relative to a reference metric. The extensions have positive tangential curvature along E; near a node use the embedding sections themselves, whose differentials span the ambient cotangent space, so the curvature is positive in both directions there. Patching can be chosen to keep these metrics near the nodes. If $q = \|s_E\|^2$ for any smooth metric on $\mathcal{O}_S(E)$, then $dd^c q$ is positive in the normal direction at every smooth point of E. Adding Cq to the weights, with $C \gg 1$, gives a metric h of positive curvature near all of E. Hence $-\log \|s_A\|_h^2$ is the required collar. For $\lambda = 0$, the positive kernel has a rational generator; clearing denominators proves (5.1). Effectiveness gives $A \cdot C \geq 0$ for every curve $C \not\subset E$, which proves nefness in the projective case. □

Boundary resolution preserves X when the new boundary is the full reduced inverse image. Interior blowups preserve both exhaustion properties by Proposition 4.1. Contracting a curve meeting but not contained in the boundary need not give a proper map of complements; no such reduction will be used.

Proposition 5.2. *A connected noncompact complex surface with a continuous psh exhaustion and $\mathcal{O}(X) \neq \mathbb{C}$ has a smooth psh exhaustion.*

Proof. Vîjîitu's theorem [4, Theorem 1.1] gives holomorphic convexity. Pull back a smooth strictly psh exhaustion of the Stein Remmert reduction under its proper reduction map. □

Proposition 5.3. *For a line bundle A on a compact Riemann surface B,*

$$\operatorname{Tot}(A^{-1}) \text{ continuously weakly complete} \iff \deg A \geq 0 \iff \operatorname{Tot}(A^{-1}) \text{ smoothly weakly complete.}$$

Proof. For $d = \deg A \geq 0$, choose an area form ω of integral one and solve $dd^c f = d\omega - \theta_0$, where θ_0 is the curvature of a reference metric. The resulting metric has curvature $d\omega$, and Lemma 3.1 applies. If $d < 0$, the infinity section of $\mathbf{P}_{\text{lines}}(\mathcal{O}_B \oplus A^{-1})$ has normal bundle A and negative square, contradicting Lemma 5.1(i). □

5.2. **Current averaging and regularization.**

Theorem 5.4. *Let S be compact Kähler and $D \neq 0$ a reduced divisor. If $X = S \setminus D$ has a continuous psh exhaustion, it has a smooth one. If X is not a modification of a Stein space, there is*

$$A = \sum_i a_i D_i, \qquad a_i \in \mathbb{Z}_{>0}, \qquad A \cdot D_i = A^2 = 0,$$

such that $\mathcal{O}_S(A)$ has a smooth Hermitian metric with zero curvature near D. Thus the entire boundary has a smooth pluriharmonic divergent collar.

Lemma 5.5. *Let (S,ω) be a compact Kähler surface and $X \subset S$ a connected noncompact open set with a continuous psh exhaustion ρ. If X is not a modification of a Stein space, there is a positive $(1,1)$-current T on S such that*

(5.2)

$$dd^c T = 0, \qquad \int_S T \wedge \omega = 1, \qquad \operatorname{supp} T \Subset X, \qquad \nu(T,x) = 0 \quad (x \in S).$$

Proof. Let Σ_X^0 be the intersection of the non-strongly-psh loci of all continuous psh exhaustions. By [5, Lemma 2.1],

$$(t_0,\infty) \subset \rho(\Sigma_X^0) \tag{5.3}$$

for some t_0. For $t > t_0$, no smooth strictly psh function exists near $K_t = \{\rho = t\}$. Indeed, if v were such a function, choose $\delta > 0$ with $\{t-\delta \le \rho \le t+\delta\}$ contained in its domain, and normalize $0 < v < 1$ there. For $0 < \eta < 1$, $0 < \varepsilon < \eta\delta/2$, set $g(s) = s + \eta|s-t|$. The function

$$\psi = \begin{cases} \max\{g(\rho), \rho + \varepsilon v\}, & |\rho - t| < \delta, \\ g(\rho), & |\rho - t| \ge \delta \end{cases}$$

is a continuous psh exhaustion, strongly psh near K_t, contrary to (5.3).

In the Banach space of continuous real $(1,1)$-forms on K_t, the linear subspace $\{dd^c f|_{K_t} : f \in C^\infty(S,\mathbb{R})\}$ is therefore disjoint from the open cone of positive definite forms. A local function extends by a cutoff, so this global formulation loses nothing. Hahn–Banach separation gives a nonzero positive functional annihilating the subspace. As a current, after mass normalization it satisfies

$$T_t \ge 0, \qquad dd^c T_t = 0, \qquad \operatorname{supp} T_t \subset K_t, \qquad \int_S T_t \wedge \omega = 1. \tag{5.4}$$

No closedness of the separated subspace is needed, since the positive cone is open.

Fix $t_0 < a < b$ and choose

$$t_{n,j} = a + \frac{j - 1/2}{n}(b-a), \qquad T_n = \frac{1}{n}\sum_{j=1}^{n} T_{t_{n,j}}.$$

These currents have mass one and support in the fixed compact shell $K = \rho^{-1}([a,b]) \Subset X$. A subsequence converges weakly to a positive dd^c-closed current T of mass one supported in K.

Fix a coordinate ball $B(x,2R) \Subset X$, and put $\beta = dd^c|z|^2$. Lelong–Jensen monotonicity for positive dd^c-closed currents [15, Appendix B, (B.1)–(B.2)] gives, uniformly in t, for $0 < r < R$,

$$r^{-2}\int_{B(x,r)} T_t \wedge \beta \le R^{-2}\int_{B(x,R)} T_t \wedge \beta \le C_R. \tag{5.5}$$

Only the levels in $[\inf_{B(x,r)}\rho, \sup_{B(x,r)}\rho]$ can contribute. Counting grid points yields

$$r^{-2}\int_{B(x,r)} T_n \wedge \beta \le C_R\left(\frac{\operatorname{osc}_{B(x,r)}\rho}{b-a}+\frac{2}{n}\right). \tag{5.6}$$

Multiply β by a cutoff equal to one on $B(x,R)$. The corresponding positive measures converge weakly, and the open-set inequality gives

$$r^{-2}\int_{B(x,r)} T\wedge\beta \le \frac{C_R}{b-a}\operatorname{osc}_{B(x,r)}\rho \longrightarrow 0 \qquad (r\downarrow 0). \tag{5.7}$$

Thus $\nu(T,x)=0$ on X; outside X, T vanishes locally. The fixed weak limit works for every ball, so no point-dependent subsequence or measurable selection is required. □

Lemma 5.6. *For a positive dd^c-closed $(1,1)$-current T on a compact Kähler surface with all Lelong numbers zero, there is a class $\alpha_T \in H^{1,1}(S,\mathbb{R})$ satisfying*

$$\alpha_T\cdot[\eta] = \int_S T\wedge\eta \qquad (d\eta=0), \tag{5.8}$$

for every smooth real $(1,1)$-form η. Moreover,

$$\alpha_T^2 \ge 0, \qquad dT\neq 0 \implies \alpha_T^2>0. \tag{5.9}$$

If $dT=0$, then $\alpha_T=[T]$.

Proof. The decomposition [15, §2.1 and Lemma B.2]

$$T=\Omega+\partial R+\bar\partial\overline{R}+i\partial\bar\partial u$$

has Ω smooth, real, and closed. Integration by parts gives (5.8) with $\alpha_T=[\Omega]$; nondegeneracy of the intersection pairing makes this class canonical. The density formula [15, Theorem 2.2(2), Corollary 2.4] gives

$$\alpha_T^2 = \|\vartheta\|+2E(T), \qquad \|\vartheta\|\ge 0, \quad E(T)\ge 0, \quad E(T)=0\Longleftrightarrow dT=0.$$

Here ϑ is the positive measure on the diagonal determined by a tangent current of $T\otimes T$, and $E(T)=\int_S \bar\partial R\wedge\partial\overline{R}$ is the current energy, independent of the chosen decomposition. Its mass hypotheses hold because the positive-Lelong-number locus is empty; no foliation hypothesis is used. This proves (5.9). □

Proof of Theorem 5.4. The Stein-modification case follows by proper pullback of a smooth psh exhaustion. Otherwise the preceding lemmas give T, supported in $K\Subset X$, and $\alpha=\alpha_T$ with

$$\alpha\cdot[\omega]=1, \qquad \alpha^2\ge 0, \qquad dT\neq 0\Longrightarrow\alpha^2>0. \tag{5.10}$$

For each component D_i, choose a metric of curvature θ_i and canonical section s_i. Cut off $\log\|s_i\|^2$ to a smooth function f_i on S agreeing with it near K. Poincaré–Lelong and dd^c-closedness give

$$\alpha\cdot[D_i]=\int_S T\wedge\theta_i=-\langle T, dd^c f_i\rangle=0. \tag{5.11}$$

For a connected boundary component E, let λ_E be the largest eigenvalue of its intersection matrix I_E. Lemma 5.1(i) excludes $\lambda_E < 0$. If $\lambda_E > 0$, rational approximation gives an integral divisor B supported on E with $B^2 > 0$. Then $\alpha \cdot [B] = 0$, and Hodge index forces $\alpha^2 < 0$, contrary to (5.10). Therefore $\lambda_E = 0$. Perron–Frobenius gives a one-dimensional positive rational kernel of I_E. Choose a positive integral generator for each E and sum:

$$A = \sum_i a_i D_i, \qquad a_i > 0, \qquad \operatorname{supp} A = D, \qquad A \cdot D_i = A^2 = 0. \tag{5.12}$$

These matrix arguments do not require smoothness or normal crossings of the components.

Since $[A] \neq 0$ is isotropic, Hodge index makes $[A]^\perp$ negative semidefinite with radical $\mathbb{R}[A]$. Using (5.10)–(5.11), we obtain

$$dT = 0, \qquad [T] = \alpha = c[A], \qquad c = \left(\int_A \omega \right)^{-1} > 0. \tag{5.13}$$

The dd^c-lemma for currents now supplies a real distribution u with

$$dd^c u = T - c[A]. \tag{5.14}$$

On a neighborhood U of D disjoint from K, a local defining function f_A gives $dd^c(u + c \log |f_A|^2) = 0$. Elliptic regularity for pluriharmonic distributions yields

$$u = -c \log |f_A|^2 + h, \qquad h \text{ smooth and pluriharmonic.} \tag{5.15}$$

Thus u is smooth pluriharmonic on $U \setminus D$ and diverges uniformly at D, including singular points, because every $a_i > 0$. Lemma 5.1(ii) gives the exhaustion.

For the metric statement, write $s_A = f_i e_i$ and set $b_i = u/c + \log |f_i|^2$ on U. These functions extend smoothly and pluriharmonically across D. If $e_j = g_{ij} e_i$, then $b_j = b_i - \log |g_{ij}|^2$; hence $\|e_i\|^2 = e^{-b_i}$ defines a flat Hermitian metric and $-\log \|s_A\|^2 = u/c$. Patch it with any global smooth metric using a cutoff equal to one on a smaller neighborhood of D. This gives the claimed global metric with curvature zero near D. □

Corollary 5.7. *Every smooth complex quasi-projective surface with a continuous psh exhaustion has a smooth psh exhaustion.*

Proof. Resolve a projective closure and principalize the complement ideal, with centers outside X [16, Theorems 1.0.1–1.0.2]. This realizes X as the complement of an SNC divisor in a smooth projective surface; the full inverse image includes any isolated boundary points. Apply Theorem 5.4. Compact components need only constant functions, and there are finitely many components. □

The case $\mathcal{O}(X) = \mathbb{C}$ occurs: take $X = \operatorname{Tot}(A^{-1})$ for non-torsion $A \in \operatorname{Pic}^0(B)$. Its squared unitary-flat norm is a smooth psh exhaustion. The

degree-n fiberwise Taylor coefficient of $f \in \mathcal{O}(X)$ lies in $H^0(B, A^n) = 0$ for $n > 0$; the constant coefficient is constant on B. Thus $\mathcal{O}(X) = \mathbb{C}$.

Corollary 5.8. *Let $C \subset S$ be a smooth connected square-zero curve in a compact Kähler surface. A continuous divergent psh collar at C implies a smooth psh exhaustion of $S \setminus C$. If the complement is not a Stein modification, $\mathcal{O}_S(C)$ is unitary flat near C and admits a pluriharmonic divergent collar.*

Proof. Take the maximum of the given collar and a sufficiently large constant near the boundary of a smaller neighborhood, and extend by that constant. This gives a continuous exhaustion. In Theorem 5.4, the single-component kernel generator can be chosen to be 1, so $A = C$. □

The theorem asserts existence of a new exhaustion, not approximation of the given one. For reducible D, flatness concerns $\mathcal{O}_S(A)$, not necessarily $\mathcal{O}_S(D)$. No divisor compactification is asserted for an arbitrary noncompact surface.

5.3. **Numerical existence criteria.**

Theorem 5.9. *For a nonprojective S and an effective $D \neq 0$, the following are equivalent:*

(i) *$S \setminus \operatorname{supp} D$ is continuously weakly complete;*
(ii) *$S \setminus \operatorname{supp} D$ is smoothly weakly complete;*
(iii) *no connected boundary component has negative definite intersection matrix.*

The Hopf case uses [11, Theorem 1.6].

Proof. Only (iii)⇒(ii) requires proof. The classification facts used below are recalled in [7, §2]. If $a(S) = 1$, the algebraic reduction is a holomorphic elliptic fibration $f : S \to B$ with connected fibers, and every curve is vertical. For a fiber $F = \sum m_i C_i$,

$$x^{\mathsf{T}} I x = -\sum_{i<j} (C_i \cdot C_j) m_i m_j \left(\frac{x_i}{m_i} - \frac{x_j}{m_j} \right)^2, \qquad I = (C_i \cdot C_j). \tag{5.16}$$

Every proper principal submatrix is negative definite. Hence the boundary contains each fiber support it meets, and $X = f^{-1}(B \setminus P)$ for a finite nonempty P. Pull back a smooth subharmonic exhaustion of $B \setminus P$.

Suppose $a(S) = 0$. By [7, Lemma 2.1], $c_1(L)^2 \leq 0$ for every line bundle L. Perron–Frobenius and (iii) therefore give an effective A of full boundary support with $A^2 = 0$. At a blowdown $q : T \to T'$, write $A = q^* A' + bE$. Then

$$0 = A^2 = (A')^2 - b^2, \qquad (A')^2 \leq 0,$$

so $b = 0$. On the minimal model $p : S \to S_0$ we obtain $A = p^* A_0 \neq 0$, $A_0^2 = 0$, and $\operatorname{supp} A = p^{-1}(\operatorname{supp} A_0)$. The map of complements is proper, so an exhaustion downstairs suffices.

If S_0 is Kähler, it is a torus or K3 of algebraic dimension zero. Such a torus has no curves: an irreducible curve would have square zero and elliptic normalization, hence be a translate of an elliptic subtorus and give a nonconstant meromorphic function by the quotient. On a K3, Riemann–Roch gives $h^0(\mathcal{O}(A_0)) \geq 2$. Both contradict $a(S_0) = 0$.

Otherwise S_0 is of class VII. If $b_2(S_0) = n > 0$, Enoki's square-zero-divisor theorem [8, Main Theorem, p. 454] makes A_0 a multiple of the distinguished cycle of an Enoki surface. Its complement is an affine-line bundle of degree $-n$ over an elliptic curve [8, (0.4), p. 453]. The infinity section of its $\mathbf{P}^1$-bundle completion has square $n > 0$, so Lemma 5.1 gives an exhaustion. If $b_2(S_0) = 0$, a minimal class VII surface containing a curve is elliptic or Hopf [8, p. 453]; $a(S_0) = 0$ leaves Hopf. Its curves are disjoint smooth elliptic curves: adjunction gives arithmetic genus one, and rational normalizations are impossible because maps $\mathbf{P}^1 \to S_0$ lift to $\mathbb{C}^2 \setminus \{0\}$. Finite Ueda type gives a smooth strictly psh collar [10, Theorem 2.5]. Infinite type gives a unitary flat neighborhood by [11, Theorem 1.6], hence a pluriharmonic collar. Glue the collars. □

Theorem 5.10. *Let $f : S \to B$ be a smooth $\mathbf{P}^1$-bundle over a compact Riemann surface and $D \neq 0$ reduced SNC. The three conditions in Theorem 5.9 are equivalent, with no restriction on $D \cdot F$.*

Proof. Necessity and positive-eigenvalue boundary components are handled by Lemma 5.1. For a connected component E with zero largest eigenvalue, choose its positive kernel divisor A. If E is vertical, it is a whole fiber, and the punctured-base construction applies. Otherwise $A \cdot F > 0$ excludes vertical components. The ruling is projective; choose a section C_0 with $C_0^2 = -e$. From $A \equiv aC_0 + bF$, $A^2 = 0$, and $A \cdot E_i = 0$ one obtains

$$b = ae/2, \qquad E_i \equiv d_i(C_0 + (e/2)F), \qquad E_i^2 = E_i \cdot E_j = 0 \quad (i \neq j).$$

Thus $E = C$ is one smooth square-zero curve. For $d = C \cdot F$, adjunction gives $2g(C) - 2 = d(2g(B) - 2)$, so $C \to B$ is finite étale by Riemann–Hurwitz.

For $d = 1$, the complement is an affine-line bundle with degree-zero linear bundle. The construction of [12, Proposition 2.1] can be written in unitary flat frames as follows. Choose a smooth section whose $\bar{\partial}$ is harmonic. Local antiholomorphic primitives g_i, followed by holomorphic translations of the affine coordinates, give

$$w_i = a_{ij} w_j + c_{ij}, \qquad g_i = a_{ij} g_j + c_{ij}, \qquad |a_{ij}| = 1,$$

with a_{ij}, c_{ij} constant. The functions $|w_i - g_i|^2$ agree; their Levi matrices in (z, w_i) are $\operatorname{diag}(|(g_i)_{\bar{z}}|^2, 1)$. They give a smooth psh exhaustion over compact B.

For $d \geq 2$, a finite étale Galois base change splits C into d disjoint square-zero sections. If $d = 2$, the complement is $L^\times$ with $\deg L = 0$, exhausted by $(\log \|v\|^2)^2$ for a unitary flat metric. If $d \geq 3$, three sections trivialize the ruling. The others are constant maps into $\mathbf{P}^1 \setminus \{0, 1, \infty\}$, so the complement

is $B' \times (\mathbf{P}^1 \setminus Q)$. Pull back an exhaustion of the punctured fiber. Averaging a nonnegative exhaustion over the finite deck group preserves properness and descends smoothly. Restrict the result to a collar at E and glue the boundary collars. □

This criterion uses the actual smooth ruling. Theorem 5.4 gives regularization on arbitrary projective compactifiers, but does not assert this numerical existence criterion for them.

5.4. **Square-zero curves.** For a smooth square-zero curve, infinite Ueda type is a formal condition. A unitary flat neighborhood requires convergent defining functions $w_i = t_{ij} w_j$, $|t_{ij}| = 1$.

Proposition 5.11. *Let $C \subset S$ be smooth, connected, and square-zero, with S compact Kähler. Put $N = N_{C/S}$, let $\rho : H_1(C, \mathbb{Z}) \to \mathrm{U}(1)$ be its unitary character, and set $K = \ker[H_1(C, \mathbb{Z}) \to H_1(S, \mathbb{Z})]$. The complement $S \setminus C$ has a smooth psh exhaustion under any of the following conditions:*

$$C \text{ has finite Ueda type or a unitary flat neighborhood};$$

$$N \text{ is torsion or Diophantine}; \qquad \rho(K) \text{ is finite};$$

$$q(S) \geq g(C); \qquad g(C) = 1.$$

Here Diophantine means $\operatorname{dist}(N^m, \mathcal{O}_C) \geq c m^{-\tau}$ for some $c, \tau > 0$ and all $m \geq 1$. Only the elliptic-curve condition uses [11, Theorem 1.6].

Proof. Finite type gives a smooth strictly psh collar [10, Theorem 2.5]; a flat neighborhood gives $-\log |w_i|^2$. In infinite type, Ueda's convergence theorem [9, Theorem 3] handles torsion or Diophantine N. If $\rho(K)$ is finite, some ρ^k factors through the image of $H_1(C, \mathbb{Z})$ in $H_1(S, \mathbb{Z})$ and extends to the latter group because U(1) is divisible. In the remaining non-torsion, infinite-type case, [10, Theorem 4.4] gives an effective E disjoint from C and cohomologous to mC. A flat metric on $\mathcal{O}_S(mC - E)$ and its canonical meromorphic section σ give the collar $-m^{-1} \log \|\sigma\|^2$.

For $q(S) \geq g(C)$, if $r : H^1(S, \mathcal{O}_S) \to H^1(C, \mathcal{O}_C)$ is not injective, [10, Proposition 2.7] makes a multiple of C a fiber, and the punctured-base construction applies. Otherwise r is an isomorphism. Hodge theory gives an isomorphism on rational first homology; since $H_1(C, \mathbb{Z})$ is torsion-free, $K = 0$. Finally, if $g(C) = 1$, [11, Theorem 1.6] rules out infinite type without a unitary flat neighborhood. Apply Lemma 5.1(ii) in all cases. □

6. Holomorphic convexity for curve fiberings

Complex spaces in this section are reduced, Hausdorff, and countable at infinity; smooth functions are understood in local embeddings. Fiber components refer to reduced supports.

Theorem 6.1. *Let X have pure dimension $n \geq 2$ and admit a continuous real-valued psh exhaustion ρ. If a holomorphic map*

$$F : X \longrightarrow \mathbb{C}^{n-1} \quad \textit{satisfies} \quad \dim F^{-1}(z) \leq 1 \qquad (z \in F(X)),$$

then X is holomorphically convex and admits a smooth psh exhaustion.

For smooth X, the convexity argument follows [17, Corollary 4.2 and pp. 463–464]. The cycle argument below allows arbitrary singularities of X.

6.1. **Compact fiber components.** The analytic dimension theorem gives

$$\dim_x F^{-1}(F(x)) \geq n - (n-1) = 1.$$

Applied on each local irreducible branch, it shows that every nonempty fiber is pure one-dimensional. Since the base is normal, F has analytic weighted fiber cycles $\mathcal{F}_z$ satisfying

$$|\mathcal{F}_z| = F^{-1}(z), \qquad \mathcal{F}_z = \sum_j m_j(z)[C_j(z)], \qquad m_j(z) \in \mathbb{N}_{>0};$$

see [18, Chapter I, §1(b), p. 816]. This assertion concerns geometric cycles and requires neither properness nor flatness of the local rings.

Lemma 6.2 (Uniform area)**.** *Fix a Hermitian metric h on X. Given compact $L \subset X$ and $z_0 \in \mathbb{C}^{n-1}$, there are a neighborhood B of z_0 and $A > 0$ such that*

$$\int_C h \leq A$$

for every compact irreducible component $C \subset L$ of a fiber over B.

Proof. Choose $\chi \in C_c^\infty(X)$ with $\chi \geq 0$ and $\chi = 1$ near L. Integration of a compactly supported continuous form is continuous on an analytic family of cycles [19, Chapter IV, Proposition 2.3.1]. Consequently, after shrinking B,

$$\int_C h = \int_{[C]} \chi h \leq \int_{\mathcal{F}_z} \chi h \leq A \qquad (z \in B).$$

□

Lemma 6.3 (Stability)**.** *Let $K \subset X$ be compact. If every irreducible component of $F^{-1}(z_0)$ meeting K is noncompact, the same holds for all fibers over a neighborhood of z_0.*

Proof. Otherwise choose $z_j \to z_0$, compact irreducible components $C_j \subset F^{-1}(z_j)$, and $x_j \in C_j \cap K$, with $x_j \to x \in K$. The maximum principle on the normalization of C_j makes $\rho|_{C_j}$ constant. Hence

$$C_j \subset L := \{\rho \leq \max_K \rho\} \Subset X.$$

Lemma 6.2 bounds their areas. Bishop compactness for cycles on complex spaces [20, Theorem 1] gives, after passing to a subsequence, a compact effective one-cycle

$$[C_j] \longrightarrow T, \qquad x \in |T| \subset F^{-1}(z_0).$$

Indeed, upper continuity of compact supports prevents the limit from losing x, and local convergence of integration currents puts its support in the central fiber. An irreducible component of $|T|$ through x is a compact analytic curve in a pure one-dimensional fiber, hence a full irreducible component of that

fiber. This contradicts the hypothesis. The cases $K = \varnothing$ or $F^{-1}(z_0) \cap K = \varnothing$ are immediate. □

6.2. **Convexity and reduction.** We use three complex-space criteria from [17, Propositions 2.2, 2.4 and Corollary 2.5]:

(i) A complex space with a continuous psh function u is holomorphically convex if and only if every $\{u < t\}$ is.
(ii) A pseudoconvex space locally holomorphically convex over a Stein base is holomorphically convex.
(iii) If $\pi : M \to S$ is holomorphic, S is Stein, and $D \subset M$ is pseudoconvex, then D is holomorphically convex provided $\overline{D}^M \to S$ is proper and a neighborhood of $\partial_M D$ has Stein reduced fibers.

In (iii), the auxiliary smooth function h constructed in the cited proof must be supplemented by an exhaustion η_D of D. Over a sufficiently small relatively compact base neighborhood, h is bounded on $\overline{D}$ and strictly psh in a boundary collar. Thus $\eta_D + h$ is a vertical exhaustion and is strictly psh above a constant: outside the collar it is bounded on a compact subset of D. This supplies the required 1-convex map, to which [17, Theorem 2.3] applies.

Lemma 6.4. *For every $\lambda \in \mathbb{R}$ and $z_0 \in \mathbb{C}^{n-1}$, there are $c > \lambda$ and a ball $B = B_r(z_0)$ such that*

$$D = F^{-1}(B) \cap \{\rho < c\} \quad \textit{and} \quad F^{-1}(B) \cap \{\rho < \lambda\}$$

are holomorphically convex.

Proof. Let S_0 be the set of constant ρ-values on compact irreducible components of $F^{-1}(z_0)$. Local finiteness of fiber components and compactness of $\{\rho \le R\}$ imply

$$\#(S_0 \cap (-\infty, R]) < \infty \qquad (R \in \mathbb{R}).$$

Choose $\lambda < a < c < b$ with $[a, b] \cap S_0 = \varnothing$. Lemma 6.3, applied to $K = \{a \le \rho \le b\}$, provides a ball $B = B_r(z_0)$ such that no compact irreducible component of a fiber over B meets K. Set

$$M = F^{-1}(B), \qquad D = M \cap \{\rho < c\}, \qquad U = M \cap \{a < \rho < b\}.$$

Then $\partial_M D \subset U$. A compact irreducible component of a reduced U-fiber would be a compact analytic curve in X, hence a full component of the corresponding full fiber meeting K. Thus all reduced U-fibers are Stein, by the Stein criterion for curves without compact irreducible components [4, §3, p. 1663].

The function

$$\eta = \max\left\{\frac{1}{c - \rho}, \frac{1}{r^2 - \|F - z_0\|^2}\right\}$$

is a continuous psh exhaustion of D: for $t > 0$,

$$\{\eta \le t\} = \{\rho \le c - 1/t\} \cap F^{-1}\{\|z - z_0\|^2 \le r^2 - 1/t\} \Subset D.$$

Moreover, $\overline{D}^M \to B$ is proper, since its inverse image over compact $T \subset B$ is closed in $\{\rho \le c\} \cap F^{-1}(T)$. Criterion (iii) makes D holomorphically convex; (i) applied to $\rho|_D$ gives the second assertion. □

Proof of Theorem 6.1. For $X(\lambda) = \{\rho < \lambda\}$, Lemma 6.4 makes $F|_{X(\lambda)}$ locally holomorphically convex over $\mathbb{C}^{n-1}$. The function $(\lambda - \rho)^{-1}$ exhausts $X(\lambda)$, so criterion (ii) makes $X(\lambda)$ holomorphically convex. Criterion (i) now gives holomorphic convexity of X. For its proper Remmert reduction $q : X \to Z$, pull back a smooth strictly psh exhaustion u of the Stein space Z:

$$u \circ q \in C^\infty(X) \cap \mathrm{PSH}(X), \qquad \{u \circ q \le t\} = q^{-1}(\{u \le t\}) \Subset X.$$

This also proves Theorem 1.4. □

Corollary 6.5 (Simultaneous contraction)**.** *The Remmert reduction $q : X \to Z$ has connected fibers, $q_*\mathcal{O}_X = \mathcal{O}_Z$, and $F = \widetilde{F} \circ q$ for a holomorphic $\widetilde{F} : Z \to \mathbb{C}^{n-1}$. Its nontrivial fibers are exactly the maximal connected unions of compact irreducible fiber curves of F.*

Proof. The sheaf identity makes every component of F descend. A nontrivial q-fiber is therefore a compact connected analytic subset of a one-dimensional F-fiber, hence a finite connected union of its compact irreducible components. Conversely, every compact irreducible curve maps to a point of Z. A connected union of such curves has constant ρ, lies in a compact sublevel, and is finite by local finiteness of the components of its F-fiber. It is consequently contained in a q-fiber, proving maximality. □

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